Chenyan Jing

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Education:

Johns Hopkins University, Baltimore, MD, USA

M.S.E. in Robotics (In Progress) Expected May 2026

Shanghai Maritime University, Sino-Dutch Mechatronics Engineering College, Shanghai, China

B.Eng in Mechanical and Electronic Engineering June 2024

Research Experience:

Research Assistant, ARCADE Lab, Johns Hopkins University, Baltimore, MD, USA

Advisor: Prof. Mathias Unberath

Mask-Guided Structural Priors for Surgical Tool Keypoint Localization

• Developed a multi-frame surgical tool keypoint localization pipeline that introduces mask-derived structural priors to refine heatmap-based predictions for tool tips and anchors.

• Designed a two-stage coarse-to-refine framework that first predicts coarse keypoint heatmaps from temporal image sequences and then refines them using prompted segmentation masks as explicit structural guidance.

• Investigated how explicit mask priors, auxiliary mask supervision, and stage-wise feature integration affect localization precision, heatmap sharpness, and structural consistency.

• Implemented end-to-end training, inference, and evaluation pipelines in PyTorch, including distributed multi-GPU training, heatmap generation, mask integration, and quantitative benchmarking.

• Built and curated annotated surgical video datasets and evaluated localization performance using L2 distance, accuracy, and recall metrics.

Selected Outcomes

• Tool tip: 8.15 px L2 error, 0.778 accuracy, 0.483 recall

• Tool anchor: 8.11 px L2 error, 0.709 accuracy, 0.579 recall

physiological motion.

Selected Projects

Predictive Retinal Motion Compensation for Robotic Subretinal Injection

• Built an iOCT and RGB-D vision pipeline for retinal motion compensation.

• Segmented retinal membranes and estimated the distance between the needle tip and retina.

• Used RANSAC filtering to remove segmentation outliers and improve localization robustness.

• Extracted tool tip and retinal target distance signals for compensation analysis.

• Benchmarked LSTM and Mamba models for retinal motion forecasting from depth signals.

Estimating 3D Human Pose Trajectories from Monocular Video

• Built a practical pipeline for recovering approximate 3D human pose trajectories from monocular RGB video by combining YOLOv8-pose for 2D keypoint detection, Depth Anything for relative depth estimation, and camera calibration for geometric alignment.

• Developed preprocessing and reconstruction steps that convert frame-wise 2D joints, depth maps, and bounding-box measurements into temporally ordered 3D pose estimates with scale recovery.

• Evaluated reconstructed trajectories against RGB+LiDAR references and analyzed practical error sources such as depth ambiguity, scale instability, occlusion, and calibration mismatch.

Temporal-Consistency Decoder for Surgical Video Generation

• Used HRNet to extract single-frame visual representations and extended the pipeline with a transformer-style decoder to incorporate temporal cues across frames.

• Improved frame-to-frame continuity in surgical video generation by decoding temporally informed representations on top of a single-frame backbone.

Place-and-Draw Task with UR5 Robot

• Implemented inverse kinematics and resolved-rate control for executing robotic drawing trajectories.

Design of Smart Insoles for Addressing Abnormal Gait

• Designed an FSR-based sensing platform with closed-loop haptic feedback for abnormal gait correction.

Joystick-Controlled Vehicle

• Developed PID-based motion control with infrared sensing and dual motor drivers to improve trajectory tracking.

Technical Skills:

Programming: Python, C++, MATLAB Frameworks: ROS2, PyTorch, OpenCV

Tools: Git, Linux, SOLIDWORKS, Arduino

Internship/Work Experience:

Newstart Robot System Technology Co., Ltd., Shandong, China November 2022 - February 2023

R&D Assistant Intern

• Assisted in the design and analysis of soft robotic gripper systems using SOLIDWORKS.

Zibo Guanzhong Industrial Design Institute, Shandong, China July 2022 - August 2022

Structural R&D Engineer Assistant Intern

• Supported engineering documentation and project risk analysis using FMEA methodologies.

Publications

• Chenyan Jing, et al. Dense Structural Priors for Sparse Functional Landmark Localization in Surgical Videos. Submitted to SPIE Medical Imaging.

• Contributed to AffordTissue, a dense affordance prediction project under review at MICCAI.

• Contributed to SAW, a surgical action world model project under review at MICCAI.

• Contributed to a retinal motion compensation project under review for an IEEE T-ASE special issue.

Patents:

• Garlic Crushing Device for Allicin Respiratory Therapy Apparatus (CN219614662U) September 2023

• Garlic Allicin Respiratory Device with Liquid Formulation (CN219517475U) August 2023

• Wearable Respiratory Mask for Garlic Allicin Respiratory Therapy (CN219517464U) August 2023

• Titration Equipment for Drug Testing and Analysis (CN219320211U) July 2023

• Food Detection Sampler (CN219284730U) June 2023